Title:

Automated and Explainable Outcome Prediction in NSCLC: Beyond Manual Feature Extraction using LLM Embeddings

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Background

In precision oncology, extracting prognostic markers from unstructured clinical notes is a labor-intensive process. Within the Apollo 11 project (NCT05550961), we developed an end-to-end pipeline that replaces manual feature engineering with Large Language Model (LLM) embeddings to predict survival and response in Non-Small Cell Lung Cancer (NSCLC).

Methods

We analyzed clinical reports from 691 patients (Fondazione IRCCS Istituto Nazionale dei Tumori, Milan). Instead of manual extraction, Qwen2.5-7B generated context-aware embeddings from textual histories. These were the sole inputs for:

1. Classification models for Overall Survival (6 and 24 months) and Disease Control Rate (DCR).
2. Survival Analysis for Overall Survival (OS) and Progression-Free Survival (PFS).

We implemented a granular explainability framework that assigns a score to every sentence in the original text, quantifying its specific impact in shifting the prediction toward a positive or negative outcome. Reliability was validated via Fidelity testing: sentences concordant with the predicted outcome (positive scores for Class 1, negative for Class 0) were concatenated and re-embedded to confirm the original prediction was maintained with high probability.

Results

This automated approach effectively predicted OS, PFS, and DCR. The explainability layer identified specific clinical drivers within the narrative, while Fidelity analysis confirmed that sentence-level scores accurately and consistently represent the model's decision-making logic.

Conclusion

LLM-based embeddings can replace manual feature extraction, streamlining real-world evidence analysis. This transparent, sentence-by-sentence breakdown provides an efficient and trustworthy decision-support tool for NSCLC.